

Exercise sheet 9

This assignment is due on March 27th at noon, to be submitted electronically via Blackboard.

1. Let $M_N = (\xi_{i,j})_{1 \leq j, k \leq N}$ be an $N \times N$ Wigner matrix where

- $\xi_{i,i} = \frac{1}{\sqrt{N}} Y_i$
- $\xi_{i,j} = \frac{1}{\sqrt{N}} Z_{i,j}$, for $i < j$
- $\xi_{j,i} = \xi_{i,j}$,

and $Y_i := P_i - 1$, with P_i all i.i.d. Poisson(1) random variables; $Z_{i,j} := B_{i,j} - 5$, with $B_{i,j}$ all i.i.d. Binomial(10, 0.5) random variables (so $Y_i, Z_{i,j}$ are centred).

- (a) If we were to plot the eigenvalues of M_N , describe the limiting shape we would expect to see including the size of the support.
- (b) In the proof of ‘step 1’ in section 17.2 of the notes, we have

$$\frac{1}{N} \mathbb{E}[\text{Tr}(M_N^k)] = \frac{1}{N} \sum_{\underline{j} \in \{1, \dots, N\}^k} \mathbb{E}[\xi_{\underline{j}}]. \quad (1)$$

Set $N = 4$ and $k = 3$.

- i. How many summands are there appearing on the right hand side of the expression?
 - ii. Choose a $\underline{j} \in \{1, \dots, 4\}^3$ and write out $\mathbb{E}[\xi_{\underline{j}}]$ by unpicking the definition of $\xi_{\underline{j}}$.
 - iii. Draw the graph and walk on the graph corresponding to your $\underline{j}$.
 - iv. Hence find $\Pi(G_{\underline{j}}, w_{\underline{j}})$, using the graph structure to justify the answer that you find.
- (c) Using (1) with $N = 4$ and $k = 2$, compute $\frac{1}{N} \mathbb{E}[\text{Tr}(M_4^2)]$.
 - (d) Compute $\mathbb{E}[Z_{1,2}^2] \cdot C_1$.
 - (e) Compare your answer to (d) and (e) in the context of Wigner’s theorem for matrix moments.
2. Let D_k denote the number of Dyck paths of length $2k$ with $k \in \mathbb{N}$.
 - (a) Setting $D_0 := 1$ show that

$$D_k = \sum_{p=1}^k D_{p-1} D_{k-p} \quad \forall k \in \mathbb{N}, \quad (2)$$

by considering the first return time of the Dyck path to zero.

(b) Define the generating function

$$D(z) := 1 + \sum_{k=1}^{\infty} D_k z^k$$

and show that it is analytic in the open disk $|z| < \frac{1}{4}$.

(c) Using (2) show that the generating function $y = D(z)$ solves the quadratic equation

$$zy^2 - y + 1 = 0. \quad (3)$$

(d) Deduce from (3) that

$$y = D(z) = \frac{1}{2z} \left(1 - (1 - 4z)^{\frac{1}{2}} \right), \quad |z| < \frac{1}{4}, \quad (4)$$

where the complex square root is chosen with cut on $[\frac{1}{4}, \infty) \subset \mathbb{R}$ such that $D(0) = 1$ (here z can be complex so we need to be careful which branch of the square-root we pick).

3. Suppose $\sigma_1(x) := \frac{1}{2\pi} \sqrt{(4 - x^2)_+}$, $x \in \mathbb{R}$ and define the Stieltjes transform of $\sigma_1(x)$ as

$$S(z) := \int_{\mathbb{R}} \frac{\sigma_1(x)}{x - z} dx, .$$

for complex $z \notin [-2, 2]$.

(a) Using the moments $m_k = \int_{\mathbb{R}} x^k \sigma_1(x) dx$ of the semicircle law and the generating function $D(z)$ in (4), show that for $z \in (0, \frac{1}{4}) \subset \mathbb{R}$,

$$D(z) = -\frac{1}{\sqrt{z}} S\left(\frac{1}{\sqrt{z}}\right). \quad (5)$$

(b) Deduce from (5) and (4) the identity

$$S(z) = -\frac{1}{2} \left(z - (z^2 - 4) \right)^{\frac{1}{2}},$$

for complex z outside of $[-2, 2]$, and where the complex square root is chosen with cut on $[-2, 2] \subset \mathbb{R}$ such that $S(z) \rightarrow 0$ as $|z| \rightarrow \infty$.